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Systematic review of female-focused attention deficit hyperactivity disorder genetics studies

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Abstract – The narrowness of the current research in attention-deficit hyperactivity disorder (ADHD) emphasizes the need for broader demographics in terms of sex and ethnic background, more specific molecular research especially in female biological systems, and to stray away from generalizations about ADHD as a behavioral disorder. Biological females may experience ADHD differently than biological males due to their different reproductive organs, hormonal pathways, and likely socialization experiences as well, which signifies the need to look more closely at the current research centering ADHD in females. This literature review will highlight what studies have been conducted that look at ADHD genetics studies exclusively in women or females. Gene-association studies found that in females with seasonal affective disorder (SAD), variation in the 5-HT2A receptor gene is associated with impulsivity and other ADHD symptoms in childhood and SAD following into adulthood. A variant of the DRD4 gene is associated with a history of childhood ADHD in probands with bulimia nervosa (BN). Understanding the genetic variation behind symptom profiles in females with ADHD may help in understanding how the behavioral disorder presents in females, and evaluating the methodology of these studies highlights considerations about trends in female-focused research, which are discussed in context of a brief review of male-focused results using the same search process.

Attention-deficit hyperactivity disorder (ADHD) is the most commonly diagnosed neurodevelopmental psychological disorder in children, with continuance into adolescence and adulthood (Cabral et al., 2020). Considering frequent issues with attention span that present even in neurotypical children, it is crucial to establish healthy and productive practices in young people to aid in learning at these important developmental stages (Francis et al., 2021). This requires understanding obstacles such as ADHD that affect learning, processing, and memory. Understanding the genetics underlying ADHD may help individuals and their families with familial genetic risk predict their needs, at school as well as at home, as well as provide insight for more individualized treatment efforts.

Although among the extensive research on ADHD a gender disparity in the prevalence of ADHD has frequently been identified, recent studies show that in adults with ADHD, the gender disparity is reduced to the point of insignificance (Attoe & Climie, 2023). More boys than girls receive treatment for ADHD, but both are similar in the genetic variability and inheritance of ADHD (Wilens et al., 2021). There are some differences in ADHD presentation identified between men and women; it is unclear if these differences are biological or social, as the literature does not tend to distinguish between sex (male vs. female) or gender (men vs. women). For this review, we will be using these terms with the acknowledgement that sex and gender are being conflated in many of these studies due to a lack of transgender and non-binary participants.

Boys receive higher scores on the scale of disruptive behavior than girls do (Slobodin & Davidovitch, 2019), but there are insufficient genetic studies of ADHD centering female populations, so perhaps disruptiveness is not a common feature in girls with ADHD. Girls with ADHD are under-diagnosed compared to boys during childhood and adolescence, and this may be because they are more likely to have symptoms of inattention instead of impulsivity and hyperactivity like boys. Oftentimes, girls will shield their symptoms to conform to standards of their parents and teachers as well as of societal norms (Mowlem et al., 2019). Males, specifically White males, are the most common participants in these genetics studies of ADHD. The narrowness of the current research in ADHD emphasizes the need for broader demographics in terms of sex and ethnic background, more specific molecular research especially in female biological systems, and to stray away from generalizations about ADHD as a behavioral disorder (Young et al., 2020).

Due to differences in biological development that become more profound at puberty, it is reasonable to investigate biological contributions to ADHD separately for males and females. Contributions of genetic variation interact with other biological factors such as hormones that differ between the sexes, so it is likely important to understand the variety of genes and biological factors involved, as well as to consider their interactions with the environment in people with ADHD. In addition to the biological sex differences that can create experiences that interact differentially with genetic factors, the socialization experiences that differ across sex and gender identities may have profound effects that also relate to genetic variation (Farhane-Medina et al., 2022). This is an area of interest for research; research using model systems has demonstrated that gene expression in the brain differs notably between the sexes even before the differentiation of the gonads (Farhane-Medina et al., 2022). This implies that from early on in growth and development, human males and females neurobiologically contrast.

Looking at genetics specifically in females is important as there is evidence that hormone-related factors may relate to ADHD, therefore genetics associated with biological differences in these hormone systems or systems that interact with them may have distinct expressions between males and females. A study examining the prevalence of symptoms of premenstrual dysphoric disorder (PMDD), episodes of postpartum depression symptoms (PPD) after the first-time giving birth, and climacteric mood symptoms in ADHD in biological females found that the fluctuating hormone levels throughout the menstrual cycle may also impact severity of ADHD symptoms (Dorani et al., 2020). This study is unique, focusing specifically on processes exclusive to biological females that may have significant interactions with ADHD diagnosis.

This literature review will highlight what studies have been conducted that look at genetic ADHD exclusively in women or females. The mechanism for genetic inheritance is the same for males and females, so aside from some differences in X-linked variants, polymorphisms have comparable prevalence. However, the impact of these genetic variants may differ based on sex or gender because of hormone levels, socialization, or experiences, thus it is important for genetics studies to make sure they are appropriately representing women and/or females. Because one of the purposes of this review is to evaluate how female-focused ADHD genetics researchers approach the studies, we did not include studies that investigate males and females simultaneously.

The purpose of this review was to highlight what studies have been conducted that investigate ADHD genetics studies exclusively in women or females. However, upon conducting the review, there is not currently sufficient published evidence for molecular genetics studies exclusively in this population. Therefore, the studies reviewed in this paper include a variety of heritability studies and genetic studies of other disorders that included ADHD comorbidities. The evidence discussed in this review emphasizes the current lack of molecular genetic studies focused on women or females.

In order to frame the amount of research, type of research, and specific findings applicable to the genetics of ADHD on female populations, we also include a review of the research on exclusively male populations to compare. Although the best way to understand any sex differences that exist in genetic contributions to ADHD is with studies including both males and females, evaluating the work that is done on the sexes separately may reveal important biases in research methodology or framing of results.

Methods

Search for Genetics of ADHD in Females

Using the PubMed database, we conducted a search on February 9th, 2023 using the terms “ADHD AND gene* AND (women OR female*)”. This yielded 7,828 results, but a majority of these articles were not focused solely on ADHD in female populations. Articles that included both male and female participants and articles that were outside the scope of the review were excluded. Inclusion criteria required that the articles included a measure of genetic variation, included an exclusively biological female population, and that these represented participants received ADHD diagnoses at a point in their lives. The variability of methods across the studies that meet the search criteria (only studies with genetics measure, ADHD diagnosis, and exclusively a female/woman population) will be discussed in the results.

Search for Genetics of ADHD in Males

For the sake of contextualizing our results of female-specific studies, we repeated the search investigating male-specific studies. Thus, using the PubMed database, we conducted a search on March 2nd, 2023 using the terms “ADHD AND gene* AND (men OR male*)”. This yielded 8,519 results, but a majority of these articles were not focused solely on ADHD in male populations either. Articles that included both male and female participants and articles that were outside the scope of the review were excluded. Inclusion criteria required that the articles included a measure of genetic

variation, included an exclusively male population, and that these represented participants received ADHD diagnoses at a point in their lives. The variability of methods across the studies that meet the search criteria (only studies with genetics measure, ADHD diagnosis, and exclusively a male/man population) will be discussed in the results and compared with the female-focused search results that are the primary focus of this review.

Results

Methodological Variation Across Female-Focused Studies

Using our primary set of search terms, six studies fit the inclusion criteria, which is that the studies must involve genetics, participants in the study must have ADHD diagnosis, and feature exclusively a female/woman population. Of these six studies, four pertain to heritability and familial risk factors (Biederman et al., 2009, Faraone et al., 2000, Faraone et al., 2001, Todd et al., 2001) and two test specific genetic variants comorbid with other psychiatric disorders (Levitan et al., 2002, Yilmaz et al., 2012).

Table 1 illustrates the sample sizes, ADHD measures/diagnoses, genetic variables, and races of participants across the six studies found that meet the inclusion criteria. All of the participants had to meet the full Diagnostic and Statistical Manual of Mental Disorders (DSM) criteria for ADHD and/or another mental disorder (Biederman et al., 2009, Faraone et al., 2000, Faraone et al., 2001, Levitan et al., 2002, Todd et al., 2001, Yilmaz et al., 2012). Most of the studies contained either all White participants (Yilmaz et al., 2012) or a majority of the participants were White (Biederman et al., 2009, Faraone et al., 2000, Todd et al., 2001). Sample sizes of participants ranged from 66 female participants (Levitan et al., 2002) to 4,036 female participants (Todd et al., 2001) and the types of genetic variable measures across these studies were familial and heritability studies (Biederman et al., 2009, Faraone et al., 2000, Faraone et al., 2001, Todd et al., 2001) as well as investigating specific genes that play a role in ADHD such as HTR2A and DRD4 (Levitan et al., 2002, Yilmaz et al., 2012).

Results Across the Studies

In a report assessing the familial transmission of ADHD in families found through girls, interviews were given to 140 girls with ADHD and their 417 first-degree relatives and to 122 girls without ADHD and their 369 first-degree relatives. The increased risk for ADHD among the relatives of girls with ADHD was true for the inattentive and combined subtypes but not for the hyperactive-impulsive subtype, which is more often seen in boys (Faraone et al., 2000). The relatives of the girls with ADHD had significantly higher rates of mood, antisocial, substance abuse, and anxiety disorders than the relatives of the girls without ADHD (Faraone et al., 2000). Performing genetic testing on relatives of girls with risk of ADHD prior to school-age could help families and individuals understand their diagnoses as well as seek any accommodations to prevent suffering and struggling at home and in school. With family members showing genetic variation of potential risk factors of ADHD, preparing for potential phenotypic outcomes could help at-risk girls with school life as well as life at home.

A study investigating whether variation in the 5-HT2A receptor gene (HTR2A) is associated with symptoms of ADHD in childhood in adult women with SAD found that in females with SAD, the variation of the 5-HT2A receptor gene plays a role in a biological process that contributes to impulsivity and core ADHD symptoms in childhood, followed by seasonal mood change in adulthood (Levitan et al., 2002).

Another study investigating the association between genetic variants and ADHD in women detected an association of the DRD4 gene with a record of attention-deficit/hyperactivity disorder in women with bulimia nervosa (BN) (Yilmaz et al., 2012). This study assessed transmission of DRD4 alleles in families of patients with BN and explored the potential impact of the DRD4 gene in childhood ADHD history in women with BN. Results showed that the BN probands and controls were not different in DRD4 allele frequency, and there was no evidence found for “preferential transmission” of any variants of DRD4 to the impacted offspring with BN. The variants of DRD4 are associated with a history of childhood ADHD in BN probands; this insight may be helpful in the treatment of BN (Yilmaz et al., 2012).

A study examining 140 girls with ADHD, 122 non-ADHD comparisons and their 786 first degree relatives aimed to test hypotheses about the familial relationship between ADHD and bipolar disorder in girls (Faraone et al., 2001). It found that relatives of both ADHD subgroups were at much higher risk for ADHD than the relatives of non-ADHD controls. A heightened risk for bipolar disorder was also found among relatives when the proband child had BPD but not ADHD by itself. There was also inadequate evidence of a discrimination between ADHD and BPD (Faraone et al., 2001).

The familiality and heritability of ADHD subtypes in a population sample of adolescent female twins was examined and latent-class analysis was used on data from parents on the DSM-IV ADHD symptoms in female twins from ages 13-23 (Todd et al., 2001). It was

Table 1
Methodology Across Studies

Citation	Sample Size	ADHD Measure/Diagnosis	Genetic Variable	Race of Participants
Biederman et al., 2009	123 probands, 403 relatives and 112 probands, 359 relatives	Met DSM-III-R diagnostic criteria for ADHD at the time of the clinical referral	Familial risk analysis	"Since the sample consisted of largely Caucasian* subjects [96% of the sample], our findings may not generalize to other minority or ethnic groups."
Faraone et al., 2000	140 girls with ADHD and their 417 first-degree relatives and to 122 girls without ADHD and their 369 first-degree relatives	Met the full DSM-III-R diagnostic criteria for ADHD at the time of the clinical referral	Family study	95% Caucasian*, 3% African American, 2% Other
Faraone et al., 2001	140 ADHD probands and 122 non-ADHD comparisons. These groups had 417 and 369 first-degree biological relatives, respectively, who provided data	Met full DSM-III-R diagnostic criteria for ADHD at the time of the clinical referral	Familiality and heritability of subtypes of ADHD	Does not say
Levitan et al., 2002	66 women	DSM-III-R for major depression/winter seasonal pattern, Wender-Utah Rating Scale for ADHD	Genomic DNA extracted, single nucleotide polymorphism (SNP) in *HTR2A* gene	Does not say
Todd et al., 2001	4,036 adolescent female twins in 1,127 monozygotic pairs and 891 dizygotic pairs	DSM-IV diagnoses were derived by computer algorithm, latent class analysis	Familiality and heritability of subtypes of ADHD	"African Americans constituted 13.3% of the total sample. The majority of the sample was European American; less than 1% were Hispanic, Asian American, Native American, or of other self-identified ancestries."
Yilmaz et al., 2012	243 women of European Caucasian descent with current or past BN purging subtype	Current or past BN purging subtype, Structural Clinical Interview for DSM-IV for Axis I Disorders, Eating Disorder Examination, Wender Utah Rating Scale	Blood lymphocyte DNA was extracted, polymerase chain reaction (PCR), looking at *DRD4* gene	All participants were European Caucasian*

Note. *Although this language is recognized as outdated and problematic by many, we include it here for accuracy in the self-identified races of the participants in the studies.

found that the primarily inattentive and combined subtypes of ADHD co-clustered within families, but the primarily hyperactive/impulsive subtype and the individual latent-class analysis subtypes did not co-cluster (Todd et al., 2001). This demonstrates that unlike the subtypes of ADHD as shown in the DSM-IV, ADHD subtypes from latent class analysis show to be independently transmitted in families. Although heritability via twin studies is typically not sex-dependent unless the trait is X-linked, heritability rates can differ between genders likely due to genetic factors interacting with socialized environmental factors that differ between genders. This can lead to amplification of genetic contribution (Ludeke et al., 2013).

In female adolescents, familial risk analysis of the association between ADHD and psychoactive substance abuse disorder (PSUD) was studied in a controlled clinical trial, as literature findings at the time could not generalize to females (Biederman et al., 2009). It found that ADHD in the proband highly increased ADHD risk in relatives independently of the comorbidity with PSUD in relatives with ADHD or not. Trends in analysis of familial risk propose that these disorders are transmitted separately (Biederman et al., 2009).

Methodological Variation Across the Male-Focused Studies

As only six articles met the inclusion criteria for the primary female-focused purpose of this review, it made sense to do another search that is exclusive to genetics studies with only male participants with ADHD diagnosis to see how many articles would meet this inclusion criteria. Only three articles met the inclusion criteria for the male-only search (Bellgrove et al., 2005, Schulz et al., 1998, Torrioli et al., 2010). Table 2 illustrates the sample sizes, ADHD measures/diagnoses, genetic variables, and races of participants across the six studies found that meet the inclusion criteria. All of the participants had to meet the full Diagnostic and Statistical Manual of Mental Disorders (DSM) for ADHD and/or another mental disorder (Bellgrove et al., 2005, Schulz et al., 1998, Torrioli et al., 2010) although the diagnostic criteria may vary across editions. Sample sizes of participants ranged from 10 male participants (Torrioli et al., 2010) to 40 male participants (Schulz et al., 1998) and the types of genetic variable measures across these studies were genetic association studies (Bellgrove et al., 2005, Schulz et al., 1998, Torrioli et al., 2010) as well as investigating specific genes/alleles that play a role in ADHD such as 5-HT and DAT1 (Bellgrove et al., 2005, Schulz et al., 1998).

Results Across the Male-Specific Studies

One of the studies found that sons of alcoholic fathers have significantly greater cortisol responses to fenfluramine (FEN) challenge procedure relative to the sons of nonalcoholic fathers Schulz et al., 1998). This discrepancy may be a result of the different mechanisms that underlie their 5-HT stimulated release, suggesting that in boys with ADHD, those at familial risk for alcoholism may be different from those who are not at risk in 5-HT function (Schulz et al., 1998). It was also discovered in another study that treatment with valproic acid improved ADHD symptoms in fragile X syndrome boys (Torrioli et al., 2010).

The final article found that the high-risk dopamine transporter genotype (DAT1) ADHD group (meaning they have a 10-repeat DAT1 allele) displayed greater response variability on the sustained attention to response task (SART) than either the low-risk DAT1 group or healthy controls, whereas the latter two groups did not differ (Bellgrove et al., 2005). The high-risk DAT1 group showed weakened spatial asymmetry, relative to the low-risk DAT1 ADHD group. The 10-repeat DAT1 allele may mediate neuropsychological impairment in ADHD (Bellgrove et al., 2005).

Differences in Female-Specific vs. Male-Specific Search Results

There were a limited number of studies investigating either female-specific or male-specific associations between ADHD and genetics. This is likely in part to a tendency for studies to include both sexes and stratify by sex if they are interested in sex-based differences. There was a wide variety of different valuable results across all of the studies, but the variability in methods and focuses of the studies was striking. The types of comorbidities that researchers focus on in the studies differ for female-specific studies compared to male-specific studies. The female-specific studies included comorbidities such as BPD (Faraone et al., 2001), SAD (Levitan et al., 2002) and BN (Yilmaz et al. 2012), which are both diagnosed at higher rates in women and may be attached to gender stereotypes in women, whereas male-specific studies included a focus on alcoholism (Torrioli et al., 2010), which is diagnosed at higher rates in men and may be associated with gender stereotypes in men.

Discussion

The studies yielded from both searches may have similar or different methodological problems based on which sex or gender the study focuses on. As much research exists as comparison studies and that is likely the best way to understand differences between male and

Table 2
Methodology Across Studies

Citation	Sample Size	ADHD Measure/Diagnosis	Genetic Variable	Race of Participants
Bellgrove et al., 2005	Twenty two right-handed ADHD participants	DSM-IV criteria for ADHD, WISC-III	Genetic association study of low-risk and high-risk DAT1 allele	Does not say
Schulz et al., 1998	40 7-to 11-year-old prepubertal boys who met DSM-III-R criteria for ADHD	DSM-III-R criteria for ADHD, Wechsler Intelligence Scale for Children–Revised (WISC-R), Wide-Range Achievement Test–Revised (WRAT-R), Family History-Research Diagnostic Criteria (FH-RDC)	Relationship of 5-HT function to aggression	30% (*n* = 12) Caucasians*, 25% (*n* = 10) African-Americans, and 45% (*n* = 18) Hispanics.
Torrioli et al., 2010	10 male patients aged between 7 and 16 years old	FXS diagnosis, diagnosis of ADHD according to the DSM-IV, WISC-R, Conner's Teacher Rating Scale, VABS—Survey Form, CGI-S ADHD, SNAP-IV—Parent Rating Scale, K-SADS PL	Fragile X syndrome is due to expansion and methylation of the CGG sequence at the 5′ UTR of the FMR1 gene	All Caucasian participants*

Note. *Although this language is recognized as outdated and problematic by many, we include it here for accuracy in the self-identified races of the participants in the studies.

female specifics, by focusing on only one sex at a time, we can discuss some of the factors associated with sex or gender that can be contributing to how ADHD is perceived in males and females. The genetic variants identified may have different impacts across sexes due to biological and social factors, specifically, relatives of girls with ADHD are at increased risk for the inattentive and combined subtypes but not for the hyperactive-impulsive subtype, which is more common in boys (Faraone et al., 2000), and also had higher rates of mood, antisocial, substance abuse, and anxiety disorders than the relatives of girls without ADHD (Faraone et al., 2000). Gene-association studies found that in females with seasonal affective disorder, variation in the 5-HT2A receptor gene affects biological processes that lead to impulsivity and other ADHD symptoms in childhood, SAD following into adulthood, and the variant of the DRD4 gene are linked to a history of childhood ADHD in probands with BN (Yilmaz et al., 2012). There was also inadequate evidence of a discrimination between ADHD and bipolar disorder (Faraone et al., 2001).

The main findings from the studies reviewed confirm the heritability of ADHD and some genetic variants associated with symptom profiles in females with ADHD. This knowledge may help in understanding how the behavioral disorder presents in females. There are certainly distinctions between the approaches in female-specific studies versus in male-specific studies. Three of the ADHD genetics studies from this search term studied females with ADHD with a comorbid disorder (Faraone et al., 2001, Levitan et al., 2002, Yilmaz et al., 2012,). These comorbidities happen to be more prevalent in this particular gender. The same general trend showed up for the studies focused on exclusively males; they chose comorbidities that are more common in men to focus on (e.g. alcoholism; Torrioli et al., 2010).

The gender binary in this review is the standard for most research on ADHD, however, sex and gender are separate constructs, and neither are strictly binary. Sex-based biological variables may be more adequately investigated based on hormone levels or other intermediate phenotypes (Martin, 2024). In terms of

gender identity, the binary completely misses the complexity of the experiences nonbinary people have. Understanding the genetics of ADHD is more complex than boys compared to girls, and this is not emphasized in the existing research. This is a new level of investigation that may help better understand ADHD as a behavioral disorder that affects all kinds of people in different ways. There is some recent evidence investigating ADHD in nonbinary folk (Cheng et al., 2022, Conway et al., 2022, Goetz 2022, Salciunas 2021), but these studies were beyond the scope of the current review; however, as more nuanced and diverse measurements of sex and gender hopefully become more prevalent in the literature, future reviews can disentangle these factors and represent a wider portion of the human population. As nonbinary and gender nonconforming people with mental illness likely face additional adversity due to their gender fluidity, it is pertinent to investigate ADHD and other mental disorders in nonbinary people.

Our inability to find any studies that directly addressed our initial focus on molecular studies of ADHD in women or females highlights the complete lack of studies of this type. We urge future researchers to fill in this gap by focusing on understanding ADHD including the crucial molecular genetic components in specifically female populations of a wide and inclusive demographic.

Importance of Studying ADHD Specifically in Females

Studying females in isolation from males allows more focus on the female bodily, mental, and endocrine changes which differ between the sexes in ways that may be critical to understand with respect to ADHD (Eng et al., 2024). There is evidence that there are sex differences in the severity and presentation of ADHD symptoms, conduct problems, and learning problems in males and females with and without clinically recognized ADHD (Mowlem et al., 2019). In females, hyperactivity/impulsivity and conduct problems were stronger predictors of clinical diagnosis and prescription of pharmacological treatment; females with ADHD may be more easily discarded in the process of ADHD diagnosis and less likely to be administered medication unless they have prominent externalizing issues (Mowlem et al., 2019). From a consecutive cohort of Child and Adolescent Psychiatric (CAP) outpatients, emotional symptoms were more common as a cause for referral to CAP among girls; for boys, prevalence of neurodevelopmental disorders was more common for reasons for referral (Klefsjo et al., 2020).

This is consistent with results from a study comparing women with and without ADHD. Results showed that women with ADHD had three times the presence of insomnia, chronic pain, ideation of suicide, sexual abuse in their childhood, and generalized anxiety disorder as women without ADHD. Women with ADHD also have two times the prevalence of substance abuse, smoking habit, depressive issues, poverty, and childhood physical abuse compared to women without ADHD (Fuller-Thompson et al., 2016). Intensity of ADHD symptoms decreases in both genders as age increases, but women with childhood ADHD are more likely than males with ADHD to have symptoms later into adulthood (Fuller-Thompson et al., 2016). It is possible that due to the differences in reproductive organs and associated hormonal pathways, ADHD is affecting males and females differently in this regard, as the endocrine system plays a large role in mood patterns of females, specifically estrogen (Roberts et al., 2018).

Changes in estrogen can affect the intensity and presentation of ADHD symptoms at different times during lifespan, such as puberty, monthly menstrual cycle, perimenopause, and menopause. A more female-centered approach to the process of ADHD diagnosis may be the key to saving many from receiving inadequate or late diagnosis. To do so, physicians may see adopting a lifespan model of care as beneficial, to model the complex transitions and changes females (and males) undergo developmentally as clinical presentations and social circumstances also change (Young et al., 2020).

Research on pharmaceutical or therapeutic intervention options for ADHD is currently working on considering variation of the 5-HT2A receptor gene, among other therapeutic targets, as it plays a role in a biological process that contributes to impulsivity and core ADHD symptoms in childhood (Nazarova et al., 2022). 5-HT2A receptor gene drug targets may help in easing symptoms of ADHD. Variants of DRD4 are associated with several neuropsychiatric diseases such as ADHD, schizophrenia, substance abuse, mood disorders, and eating disorders such as BN (Botticelli et al., 2020), so looking more into the findings from these studies may provide researchers with important neuronal and hormonal pathways that affect ADHD in all people, not specific to a particular race or gender binary.

The approach to the current review was valuable, including providing insight into research approaches in ADHD genetic studies focusing on just one gender or sex, but the tradeoff was not including many of the excellent studies that compare gender differences by including both males and females. The limitations of this literature review include the fact that since this review only included articles exclusively with female participants, some key evidence may have been missed regarding the genetics of ADHD in females. Studies that focus on sex

differences by comparing results in both males and females are likely to be even more informative about the unique qualities of ADHD in females than those that study females alone. Many studies include both boys and girls but might do an analytic strategy that looks at association separately, thus allowing for further clarification about differences that may occur between the sexes.

The limitations of the studies in this literature review are that in the majority of the studies cited, the participants included in the studies using the specific search term were of European descent (Biederman et al., 2009, Faraone et al., 2000, Todd et al., 2001, Yilmaz et al., 2012) unless the authors did not describe racial or ethnic demographics (Faraone et al., 2001, Levitan et al., 2002). This means that conclusions about ADHD from these types of studies are still limited and narrow, as these studies include samples that do not adequately represent every person affected by ADHD and therefore cannot be generalized to all females with ADHD. Another limitation is that many of the studies found through the search term are from at least a decade ago, so some of the findings may not be as up to date with current knowledge.

Future research should represent people with ADHD who are from a variety of ethnic backgrounds, genders, races, and socioeconomic experience so that the conclusions drawn from the studies explored in this literature review can be generalized to non-White gender-varying populations. The way researchers focus on ADHD in females in these studies (and the way the researchers focused on ADHD in males) reveals how attitudes about gender shape research and limit our understanding of disorders like ADHD. It may be useful in a future review to include all studies that include gender differences, to provide a broader perspective for how researchers examine sex and gender differences in ADHD genetics. Results from the female-only studies draw suggestions for how we should do genetics studies of ADHD in females to be inclusive of females who have ADHD, taking into account the processes that are specific to biological females, such as menstruation, gestation, and menopause, as well as life experiences that are impacted by their societies and environments.

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